Binary Classification with Support Vector Machines
Assignment 2 Report

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Introduction

In this report we study and implement Support Vector Machines (SVMs) for a binary classification problem.

An SVM is a linear classifier that separates two classes with a hyperplane chosen so that the margin is as large as possible. We implement both the hard-margin and soft-margin SVM, and for each we solve both the primal and the dual optimization problems.

The decision function we use throughout follows the defined convention

$$f(x) = w^\top x - b, \qquad \hat{y} = \operatorname{sign}\left(f(x)\right), \tag{1}$$

where w is the weight vector and b is the bias (offset).

This report covers our understanding of the theory (hard- and soft-margin primal/dual forms, support vectors, the maximum-margin principle, generalization, and duality) and an experimental comparison of our implementation against libsvm on the Wisconsin Diagnostic Breast Cancer dataset.

1 Dataset and preprocessing

We use the *Wisconsin Diagnostic Breast Cancer (WDBC)* dataset. The features are computed from digitized images of fine-needle aspirates of breast masses and describe properties of the cell nuclei.

The dataset contains:

- 569 samples
- 27 real-valued input features (mean, standard error, and "worst" values of nuclear measurements such as radius, texture, perimeter, area, etc.)
- Two classes: malignant (`M`) and benign (`B`), with no missing values.

We map the labels to $\{-1, +1\}$:

$$\texttt{M} \mapsto +1 \text{ (malignant)}, \qquad \texttt{B} \mapsto -1 \text{ (benign)}. \tag{2}$$

The class balance is 212 malignant ($+1$) and 357 benign (-1).

For preprocessing, the data was split into a training set and a test set (75 % / 25 %, stratified by class). Then each feature is standardized to zero mean and unit variance, computing the mean and standard deviation on the training set only and applying the same transform to the test set.

Standardization matters because the raw features live on very different scales (e.g. area in the hundreds vs. smoothness near 0.1), and an SVM penalizes $\|w\|$ uniformly across coordinates; without scaling, the large-magnitude features would dominate and the QP would be poorly conditioned.

2 Hard-margin SVM

2.1 Primal form

When the two classes are linearly separable, the hard-margin SVM looks for the separating hyperplane with the largest margin. The margin width is inversely proportional to $\|w\|$, so maximizing the margin is equivalent to minimizing $\frac{1}{2}\|w\|^2$. The primal problem is

$$\min_{w,\,b} \ \frac{1}{2}\|w\|_2^2 \qquad \text{s.t.} \qquad y_i(w^\top x_i - b) \geq 1, \quad i = 1, \ldots, N. \tag{3}$$

Each constraint forces every training point to lie on the correct side of the hyperplane and at least a functional margin of 1 away from it. "Hard" means no point is allowed to violate its constraint.

2.2 Dual form

Introducing Lagrange multipliers $\alpha_i \geq 0$ for the inequality constraints and eliminating w and b yields the dual problem

$$\max_{\alpha} \ \sum_{i=1}^{N} \alpha_i - \frac{1}{2}\sum_{i=1}^{N}\sum_{j=1}^{N} \alpha_i \alpha_j \, y_i y_j \, x_i^\top x_j \qquad \text{s.t.} \qquad \alpha_i \geq 0, \quad \sum_{i=1}^{N} \alpha_i y_i = 0. \tag{4}$$

From the optimal α we recover the weight vector as

$$w = \sum_{i=1}^{N} \alpha_i y_i x_i, \tag{5}$$

and we estimate b from the support vectors (Section 6.2). The dual depends on the data only through the inner products $x_i^\top x_j$, which is what later allows kernels; here we use the linear kernel.

2.3 Support vectors

The Karush–Kuhn–Tucker (KKT) conditions imply that $\alpha_i > 0$ only for the training points lying exactly on the margin, i.e. for which $y_i(w^\top x_i - b) = 1$. These points are the support vectors. All other points have $\alpha_i = 0$ and do not affect the solution.

Consequently the hyperplane is determined entirely by a small subset of the data : in the experiments only 26 of the training points are support vectors for the hard-margin model. This sparsity in the training data is one of the most important properties of the SVM.

3 Soft-margin SVM

3.1 Primal form

Real data is rarely perfectly separable, and even when it is, a hard margin can overfit. The soft-margin SVM introduces non-negative slack variables ζ_i that allow some points to violate

the margin, at a cost. Following the assignment, we use the modified objective with the penalty scaled by $1/N$:

$$\min_{w,\,b,\,\zeta} \ \frac{1}{2}\,\|w\|_2^2 + \frac{C}{N}\sum_{i=1}^{N}\zeta_i \qquad \text{s.t.} \qquad y_i(w^\top x_i - b) \geq 1 - \zeta_i, \quad \zeta_i \geq 0. \tag{6}$$

3.2 Role of the slack variables

Each slack ζ_i measures how far point i is allowed to fall short of the margin: $\zeta_i = 0$ means the point satisfies the hard-margin constraint, $0 < \zeta_i \leq 1$ means it is inside the margin but still correctly classified, and $\zeta_i > 1$ means it is misclassified. The term $\frac{C}{N}\sum_i \zeta_i$ penalizes the total violation.

The parameter C controls the trade-off: a large C pushes toward the hard margin (few violations, small margin), while a small C tolerates more violations in exchange for a wider, smoother margin. Dividing by N makes the penalty an average slack per sample, so the meaning of a given C is less sensitive to dataset size.

3.3 Dual form

The corresponding dual is identical to the hard-margin dual except that the multipliers are now bounded above:

$$\max_{\alpha} \ \sum_{i=1}^{N}\alpha_i - \frac{1}{2}\sum_{i=1}^{N}\sum_{j=1}^{N}\alpha_i\alpha_j\, y_i y_j\, x_i^\top x_j \qquad \text{s.t.} \qquad 0 \leq \alpha_i \leq \frac{C}{N}, \quad \sum_{i=1}^{N}\alpha_i y_i = 0. \tag{7}$$

The upper bound is C/N (not C) precisely because the primal penalty is $\frac{C}{N}\sum_i \zeta_i$.

We again recover $w = \sum_i \alpha_i y_i x_i$.

Here the support vectors split into two kinds: points with $0 < \alpha_i < C/N$ lie exactly on the margin ("free" support vectors), while points with $\alpha_i = C/N$ are inside the margin or misclassified ("bounded" support vectors).

We estimate b from the free support vectors, where the margin constraint is tight.

4 Duality

4.1 Weak duality

For any optimization problem, the dual objective at a dual-feasible point is a lower bound on the primal objective at any primal-feasible point. Writing $p^\star$ for the optimal primal value and $d^\star$ for the optimal dual value, weak duality states

$$d^\star \leq p^\star. \tag{8}$$

This always holds, regardless of convexity.

4.2 Strong duality

When the problem is convex and a constraint qualification (such as Slater's condition) holds — which is the case for the SVM QP — the two optima coincide:

$$d^\star = p^\star. \tag{9}$$

This is why solving the dual gives exactly the same classifier as solving the primal, and it justifies recovering w and b from the dual multipliers.

4.3 Duality gap

The duality gap is the difference between the primal and dual objective values for a given pair of solutions,

$$\text{gap} = p^\star - d^\star \geq 0. \tag{10}$$

Under strong duality the gap is zero at the optimum. In practice an iterative solver stops at a slightly suboptimal point, so the measured gap is a small positive number rather than exactly zero; it is a convenient indicator of how well the solver converged. We report it for all dual experiments in Section 7.

5 Max-margin intuition and generalization

Among all hyperplanes that separate the two classes, the maximum-margin hyperplane sits as far as possible from both classes. Intuitively, a wide margin leaves the most room for new, unseen points to fall on the correct side, so the classifier is more robust to small perturbations in the data. A boundary squeezed right up against the training points is fragile; a wide-margin boundary is not.

The quantity we actually care about is the generalization error, not the training error. A classifier can achieve zero training error and still generalize poorly (overfitting). We approximate the generalization error empirically by the test-set accuracy.

Statistical learning theory explains why a large margin helps. Informally, the capacity (e.g. the VC dimension or, more tightly, the fat-shattering dimension) of the set of large-margin separators is small even in a high-dimensional space: a margin of γ on data contained in a ball of radius R limits the effective capacity by a factor on the order of R^2/γ^2, independent of the ambient dimension. The resulting bounds say, roughly,

$$\text{test error} \;\lesssim\; \text{train error} + O\left(\sqrt{\frac{R^2/\gamma^2}{N}}\right), \tag{11}$$

so a larger margin γ (smaller $\|w\|$) tightens the bound. This is the theoretical counterpart to the intuition above: maximizing the margin minimizes a bound on the generalization error, which is exactly what the SVM objective does.

6 Implementation details

We implement the four variants in a single Python file, `svm_assignment.py`, using `numpy`, `scipy`, `pandas`, and `scikit-learn`.

The custom SVMs are solved with `scipy.optimize.minimize` (SLSQP); `sklearn.svm.SVC` is used only for the libsvm comparison.

6.1 Solving the primal and the dual

Primal (hard margin)

We optimize over the stacked variable $z = [\, w\,(d);\ b\,(1)\,]$ and minimize $\frac{1}{2}\,\|w\|^2$ subject to the linear constraints $y_i(w^\top x_i - b) \geq 1$, encoded as a `LinearConstraint`.

Primal (soft margin)

We extend the variable to $z = [\, w\,(d);\ b\,(1);\ \zeta\,(N)\,]$ and minimize $\frac{1}{2}\,\|w\|^2 + \frac{C}{N}\sum_i \zeta_i$ subject to $y_i(w^\top x_i - b) + \zeta_i \geq 1$ and $\zeta_i \geq 0$.
A feasible starting point ($w = 0$, $b = 0$, $\zeta = 1$) is supplied.

Dual (hard and soft)

We build the matrix $Q_{ij} = y_i y_j\, x_i^\top x_j$ (symmetrized for numerical safety) and minimize $\frac{1}{2}\alpha^\top Q\alpha - \mathbf{1}^\top\alpha$, the negative of the dual objective, subject to the box constraints ($0 \leq \alpha_i$ for hard margin, $0 \leq \alpha_i \leq C/N$ for soft margin) and the equality constraint $\sum_i \alpha_i y_i = 0$. We pass the analytic gradient $Q\alpha - \mathbf{1}$ to the solver.

6.2 Recovering w, b, and α

From a dual solution we set $w = \sum_i \alpha_i y_i x_i$.

To recover b we use the KKT condition that a free support vector lies exactly on the margin, $y_i(w^\top x_i - b) = 1$, which gives $b = w^\top x_i - y_i$.

We average this over all free support vectors (those with $0 < \alpha_i < C/N$ in the soft case, or all support vectors in the hard case) for numerical stability.

In the primal solvers, w and b are read directly from the optimized variable vector, and the support vectors are identified as the points lying on or inside the margin.

6.3 Sign convention for b

The assignment uses $f(x) = w^\top x - b$, whereas libsvm / sklearn use $f(x) = w^\top x + b_{\text{sklearn}}$ (i.e. `coef_` and `intercept_`).
The two describe the same hyperplane provided

$$b = -\,b_{\text{sklearn}} = -\,\texttt{intercept_}. \tag{12}$$

We apply this sign flip whenever we compare our b with sklearn's, and we keep the $w^\top x - b$ form consistently throughout our own code so that the decision function and the bias estimate agree.

6.4 Comparison with libsvm / sklearn

Two conventions must be reconciled. First, the sign of the bias, handled above. Second, the regularization scaling: our primal penalty is $\frac{C}{N}\sum_i \zeta_i$, while libsvm uses $C_{\text{libsvm}}\sum_i \zeta_i$ with no $1/N$ factor. To make the two objectives identical we therefore set

$$C_{\text{libsvm}} = \frac{C}{N}. \tag{13}$$

For the hard-margin comparison we approximate the hard margin with a very large C. We then extract `coef_` and `intercept_` from the fitted `SVC`, convert to our $w^\top x - b$ convention, and compare w, b, the dual coefficients α, the decision scores on the test set, and the test accuracy.

6.5 Numerical tolerances

Small differences between our solver and libsvm are expected and come from three sources:

1. *Optimization tolerance* : SLSQP and libsvm stop at slightly different points within their convergence thresholds

2. *Feature scaling and conditioning* : the hard-margin problem in particular is ill-conditioned, which amplifies tiny differences

3. *Different handling of the bias / intercept* : the two libraries estimate b from possibly different sets of support vectors. We therefore compare using norms and correlations rather than expecting bit-for-bit equality, and we treat agreement at the level of 10^{-4}–10^{-6} as confirmation of correctness.

7 Experimental results

We run all four variants with $C = 1.0$, a 25 % stratified test split, and random seed 42. The training set has $N = 426$ samples, so $C/N \approx 2.35 \times 10^{-3}$. All values below are taken directly from the program output.

7.1 Accuracy and support vectors

Model	Train acc.	Test acc.	#SV	$\|w\|$	b	Solver
hard-primal	1.0000	0.9091	26	113.7819	−5.1367	OK
hard-dual	1.0000	0.9091	26	113.7817	−5.1371	WARN
soft-primal	0.9507	0.9510	162	0.4250	0.3416	OK
soft-dual	0.9507	0.9510	162	0.4250	0.3417	OK

The hard-margin models reach 100 % training accuracy, confirming that the standardized WDBC data is linearly separable with all 27 features. However, their test accuracy is lower (0.9091) and $\|w\|$ is very large (≈ 113.8): forcing every point outside the margin squeezes the margin and inflates the weights, a sign of mild overfitting. The soft-margin models accept a small training error (train accuracy 0.9507) but generalize slightly better (test accuracy 0.9510) with a far smaller $\|w\| \approx 0.425$ — a wider, smoother margin. This is the maximum-margin / regularization trade-off in action. The hard-margin models use only 26 support vectors, while the soft-margin models use 162 (many of them bounded support vectors inside the wider margin).

7.2 Objective values and duality gap

Model	Primal obj.	Dual obj.	Duality gap
hard-primal	6473.1588	—	—
hard-dual	6473.1405	6473.1588	-1.83×10^{-2}
soft-primal	0.282966	—	—
soft-dual	0.282967	0.282966	1.83×10^{-7}

The soft-margin dual shows a duality gap of about 1.8×10^{-7}, essentially zero, a clean confirmation of strong duality: the primal and dual reach the same optimal value. The hard-margin dual reports a small negative gap (-1.8×10^{-2}) and the solver emits a `WARN` ("Positive directional derivative for linesearch"). A negative gap is impossible analytically (weak duality requires gap ≥ 0); it appears only because the hard-margin dual is ill-conditioned and SLSQP stopped slightly short of the true optimum, so its reported dual value marginally exceeds the primal. This is a numerical-tolerance artifact, not a modeling error: the resulting w, b, accuracy, and support-vector count all match the cleanly solved hard-primal model and libsvm.

7.3 Comparison with libsvm

For each model we compare against `SVC` with $C_{\text{libsvm}} = C/N$ (and a large C for the hard-margin approximation), after applying the $b = -$`intercept_` sign flip.

Model	$\|\Delta w\|$	$\|\Delta b\|$	$\|\Delta \alpha\|$	$\max \|\Delta f\|$	$\text{corr}(f)$	Acc. ours / libsvm
hard-primal	2.56×10^{-2}	1.63×10^{-3}	—	5.20×10^{-2}	1.000000	0.9091 / 0.9091
hard-dual	2.60×10^{-2}	1.15×10^{-3}	7.53×10^{-1}	5.06×10^{-2}	1.000000	0.9091 / 0.9091
soft-primal	1.23×10^{-4}	5.01×10^{-5}	—	5.18×10^{-4}	1.000000	0.9510 / 0.9510
soft-dual	5.57×10^{-6}	1.04×10^{-5}	1.22×10^{-6}	2.31×10^{-5}	1.000000	0.9510 / 0.9510

The correlation between our decision scores and libsvm's is 1.000000 for all four models, and the test accuracies are identical. This is the strongest evidence of correctness: our hyperplane points in exactly the same direction as libsvm's, up to numerical precision. The soft-margin models agree extremely tightly ($\|\Delta w\| \sim 10^{-4}$–$10^{-6}$, and $\|\Delta \alpha\| \sim 10^{-6}$ for the dual), confirming that the $C_{\text{libsvm}} = C/N$ correspondence is correct.

The hard-margin differences are larger (e.g. $\|\Delta w\| \approx 2.6 \times 10^{-2}$, and $\|\Delta \alpha\| \approx 0.75$ for the dual). Two reasons: the hard margin is approximated by a finite large C on the libsvm side, and the dual α are not uniquely pinned down the same way when many constraints are active and the problem is ill-conditioned. Since the recovered w, the object the α reconstruct, still matches and the correlation is 1, the two models are equivalent in practice.

7.4 Primal–dual consistency

As an internal check we compare the primal and dual solutions of the same model directly:

	$\|w_{\text{primal}} - w_{\text{dual}}\|$	$\|b_{\text{primal}} - b_{\text{dual}}\|$
hard	2.66×10^{-3}	4.79×10^{-4}
soft	1.24×10^{-4}	3.97×10^{-5}

The primal and dual produce the same parameters up to small numerical differences, as strong duality predicts.

7.5 Confusion matrix

On the test set, the soft-margin models give the confusion matrix (rows = true $[-1, +1]$, columns = predicted $[-1, +1]$)

$$\begin{bmatrix} 90 & 0 \\ 7 & 46 \end{bmatrix}, \tag{14}$$

i.e. all benign cases are classified correctly and 7 malignant cases are missed. In a medical setting the false negatives (malignant predicted benign) are the more costly errors, which one could address by adjusting the decision threshold or the class weighting, a possible extension.

8 Conclusion

We implemented hard- and soft-margin SVMs in both their primal and dual forms, solving each as a quadratic program with a generic optimizer and without calling any SVM library for the classifier itself. On the WDBC dataset, the hard-margin model separates the training data perfectly but generalizes slightly worse, while the soft-margin model trades a small training

error for a wider margin and marginally better test accuracy (0.9510), illustrating the maximum-margin and regularization principles. All four variants match the libsvm reference with decision-score correlation 1.000000 and identical test accuracy; the soft-margin parameters agree to within $\sim 10^{-6}$ once the $b = -\texttt{intercept_}$ sign convention and the $C_{\text{libsvm}} = C/N$ scaling are accounted for. The soft-margin dual exhibits a duality gap of $\sim 10^{-7}$, confirming strong duality, while the small negative gap and solver warning in the hard-margin dual are explained as numerical artifacts of an ill-conditioned problem rather than modeling errors. Overall, the experiments confirm that our implementation is correct and behaves consistently with the theory and with libsvm.